

The total biomass nitrogen reservoir and its potential of replacing chemical fertilizers in China

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ABSTRACT

Biomass resources originally produced by photosynthesis and their biodegradable derivatives such as crop straw, vegetable waste, livestock and poultry manure, etc., can be used as organic fertilizers which have huge potentials in replacing chemical ones. However, it is not clear how large the biomass nitrogen reservoir is in a certain region or country. In order to determine the biomass nitrogen reservoir and its potential to replace chemical fertilizers, we took China as a case, investigated 12 kinds of biomass resources and calculated their nitrogen reservoirs. It was found that the total amount of biomass (dry weight) in China was about 1.527 billion t·a⁻¹ and the urine of animal (wet weight) was about 878.75 million t·a⁻¹. The sum of them contained 25.53 million t·a⁻¹ nitrogen, being 4.12 times that of the amount of agriculture plants actual absorption for synthetic nitrogen fertilizers (6.20 million t·a⁻¹) in the country in 2018. For biomass nitrogen, the biggest contributor came from the manure and urine of livestock and poultry (54.5%), followed by crop straw (22.9%), vegetable waste (9.4%), vinasse, vinegar and soy sauce lees (2.3%), municipal sludge (2.2%), garden waste (2.1%), urban and rural household garbage (1.8%), orchard waste (1.8%), fishery waste (1.6%), Chinese herbal dreg (0.64%), mushroom stick (0.63%), with slaughterhouse waste being 0.13%. We suggest the government should pay more attention to the utilization of biomass nitrogen by returning waste biological resources to agricultural ecosystem and achieve sustainable agricultural development in the future.

1. Introduction

Agriculture largely depends on fertilizers, especially chemical fertilizers to increase crop yield, with the amount being increased year by year. According to statistics, in 2016, the consumption of chemical fertilizers was 197.5 million t in the world, including 110 million t nitrogen fertilizers [1]. China is the world's largest chemical fertilizer user, consuming 35% of the world chemical fertilizers [2]. From 1980 to 2015, the amount of chemical fertilizers application in China increased from 12.69 million t·a⁻¹ to 60.23 million t·a⁻¹, with the synthetic nitrogen fertilizer (SNF) increasing from 9.34 million t·a⁻¹ to 23.62 million t·a⁻¹ [3]. However, the nitrogen utilization efficiency has been decreasing year by year since 1985 [4]. The average nitrogen utilization

efficiencies in Chinese croplands were 28.3%, 28.2% and 26.1%, respectively for rice, wheat and maize [5], far lower than those of developed countries (50%–60%) [6]. Excessive application of chemical fertilizers has also led to a series of environmental problems, such as ammonia volatilization, denitrification, green gas emission, and air pollution [7]. It further accelerates the process of global warming, soil acidification, and degradation of quality of agricultural products [8].

In a natural ecosystem, all available nitrogen comes from the air which contains nitrogen as high as 78%. Naturally, biomass nitrogen can be provided by biological nitrogen fixation, lightning nitrogen fixation, dry and wet deposition [9]. Apart from chemical fertilizer factories, human being can balance the nitrogen losses by returning straws to the field [10], planting green-manure [11], using weed [12] and applying

Abbreviations: SNF, synthetic nitrogen fertilizer; ASN, absorbed synthetic nitrogen by agriculture plants; BN, biomass nitrogen; CHD, Chinese herbal dreg; CNKI, Chinese National Knowledge Infrastructure; CS, crop straw; FW, fishery waste; GW, garden waste; HG, urban and rural household garbage; LPM, livestock and poultry manure; LPMU, manure and urine of livestock and poultry; LPU, livestock and poultry urine; MRS, mushroom stick; MS, municipal sludge; OW, orchard waste; SW, slaughterhouse waste; VSL, vinasse, sauce and soy vinegar lees; VW, vegetable waste.

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organic fertilizers such as human or animal manures. China's farmland had experienced an eco-agriculture history of over 5000 years without degradation, thanks to the wise utilization of organic fertilizers. However, historically, the lack of both labor force and organic fertilizer sources led to the low production of farmlands. Now, these problems have been basically solved through technology improvement on cultivating soil carbon and maintaining nitrogen reservoirs [13].

Compared to chemical fertilizers, organic ones have higher organic matter, more comprehensive nutrients and longer fertilizer effects, which could improve the structure of soil microbial community and soil maintenance capacity, increase nitrogen utilization efficiency, soil fertility and finally the quality of agricultural products. Traditional organic fertilizers have been produced by composting human, livestock and poultry manure (LPM) and crop straw (CS) together [14]. Previously, organic fertilizers were only limited to human excrement and urine or animal manure, now they have been expanded to all organic matters from all ecosystems, such as farmlands, orchards, vegetables, gardens, urban and rural ecosystems or some natural ones [13].

CS is one of the main products in an agricultural ecosystem. China is the largest agricultural country in the world, with a wide variety of straw resources. Returning of straw to the field could not only increase soil organic matter content, but also improve soil structure [15]. So the CS is a kind of ideal biomass fertilizer to the farmlands. However, most of the CS resources are wasted in China which seriously contaminate the environment due to the lack of practical treatment and utilization technologies. The returning percentage of CS is much lower than Europe, the United States and other developed countries [16].

Livestock and poultry industries which provide human being with a basic source of daily protein contribute nearly half of the world's agricultural economy [17]. Unfortunately, LPM emission has become one of the most serious environmental pollutants other than utilized as organic fertilizers [18]. China is also the largest livestock and poultry breeding country, with a huge amount of LPM being produced every year. According to the Statistic of the Ministry of Agriculture in 2016, the sewage of livestock and poultry in China had reached to 3.80 billion t·a⁻¹, however, the comprehensive utilization rate is less than 60% [19]. How to solve the LPM pollution and effectively increase the bio-resource utilization has become an urgent task.

Although some researchers investigated biomass resource distributions in previous studies (Table 1), they mainly focused on CS, LPM, household waste and other rural waste resources. So the previous research object was single and the scope was small. As there is no

systematic research on the total biomass nitrogen reservoir in China, so we still do not know how much SNF could be replaced by the biomass nitrogen resources.

It is well known that all biomass is produced by photosynthesis and its biodegradable derivatives can be used as raw materials for organic fertilizer production. In addition to the above mentioned biomass resources, materials from municipal sludge; wastes from orchard, vegetable and garden; lees from productions of wine, soy sauce and vinegar; Chinese herbal dreg; mushroom stick; slaughterhouse waste *etc.*, should be all included as organic fertilizer materials. By doing so, chemical fertilizers might be significantly reduced, thus to maintain nitrogen balance, slow down environmental pollution and promote sustainable development of agriculture.

Therefore, the scientific hypothesis of this paper is that the total nitrogen amount in agricultural and non-agricultural systems might reach or exceed the amount of SNF. The utilization of such renewable nitrogen resources can greatly reduce environmental pollution and improve soil fertility. The aims of this study are to try to reveal: 1) How much biomass resources are distributed in China; 2) To what extent can the SNF be potentially replaced by biomass nitrogen resources in China. This study might provide an important theoretical basis for chemical fertilizer reduction and eco-agriculture development.

2. Methodology

2.1. Information sources

According to the theory that all biomass resources originally produced by photosynthesis and their biodegradable derivatives can be used as raw materials for organic fertilizer production, here, 12 kinds of main biomass resources, *i.e.*, crop straw, the manure and urine of livestock and poultry, urban and rural household garbage, municipal sludge, slaughterhouse waste, fishery waste, vegetable waste, garden waste, orchard waste, vinasse, vinegar and soy sauce lees, Chinese herbal dreg, and mushroom stick were selected to calculate their nitrogen reservoirs.

The original data of biomass resources mainly came from: China Statistical Yearbook (2010–2019); China Rural Statistical Yearbook (2010–2019); China Statistical Yearbook on Urban and Rural Construction (2009–2018); China Fishery Statistics Yearbook (2019) and the related information obtained by searching “biomass”, “livestock and poultry manure”, “household garbage”, “municipal sludge”, “waste” and

Table 1
The research topic of relevant literatures.

The article title	The first author	Published year	The main research topic
Distribution, utilization structure and potential of biomass resources in rural China: With special references of crop residues	Liu [20]	2007	Analysis of distribution, utilization structure and energy utilization potential of CS in rural China
Nutrient resource quantity of crop straw and its potential of substituting	Song [21]	2018	The CS and nutrient resources
Research on the resource utilization of organic waste in rural areas	Ding [22]	2015	The rural straw, livestock manure, domestic waste resources
Straw utilization in China—status and recommendations	Ren [23]	2019	The quantity and utilization of straw resources
Studies on the evaluation of nutrient resources derived from manure and optimized utilization in arable land of China	Jia [24]	2014	The resource quantity and nutrient content of straw and livestock manure in China
Organic fertilizer resources and utilization in China	Niu [25]	2017	The resources and utilization of organic fertilizer in China over the past 30 years including manure, urine, straw and green manure
The utilization potential of the main organic nutrient resources in China	Hou [26]	2017	The CS and livestock manure resources and nutrient content
Study on nutrients balance and requirement in agricultural production in China	Liu [27]	2018	The resource quantity and nutrient content of straw and livestock manure in China
Domestic solid waste discharge and its determinants in rural China	Huang [28]	2014	The rural domestic waste resources in China
A comprehensive overview of rural solid waste management in China	Zeng [29]	2015	The utilization, characteristics and management of rural solid waste in China
Progress of rural solid waste resource utilization in China	Min [30]	2020	The amount of straw, livestock excrement, domestic waste and its utilization in China's rural areas
Research status and prospects of the resource utilization of organic waste in urban and rural areas	Li [31]	2019	The utilization and research status of the resources such as food and kitchen waste and municipal sludge

other keywords through the website of “Web of Science” and “Chinese National Knowledge Infrastructure (CNKI)”.

2.2. Main crop straw and nitrogen contents

Maize, wheat, rice, beans, tubers, cotton, oil crops, sugar crops and hemp crops were investigated according to the planting structure of major crops in China. Crop yield data were selected from China Statistical Yearbook (2010–2019) [3]. The calculation methods of straw biomass and nitrogen resource were referred to relevant literature [32]. The specific calculation equations are as follows:

$$Q_j = \sum_{n=1}^j A_n \times k_n \quad (1)$$

$$N_j = \sum_{n=1}^j Q_n \times \alpha_n \quad (2)$$

where Q_j is the total amount of straw resources generated by j types crops each year; j is the total number of crop types; A_n and k_n are the economic yield and corresponding straw coefficient of n -type crop, respectively; N_j is the nitrogen amount of j types crop straws; Q_n and α_n are the straw resources and nitrogen content of n -type crop, respectively.

2.3. Manure and urine of livestock and poultry and nitrogen contents

The manure and urine of livestock and poultry (LPMU) were mainly from husbandry industries of pig, cattle, cow, sheep, horse, donkey, mule, rabbit and poultry. The calculation methods of LPMU were referred to relevant literature [27], with some modifications being as follows:

$$W_j = \sum_{n=1}^j N_n \times t_n \times R_n \quad (3)$$

$$N_j = \sum_{n=1}^j W_n \times \alpha_n \quad (4)$$

where W_j is the total output of j types animal manure each year; j is the total number of animal species; N_n , t_n and R_n are the breeding quantity, feeding period and daily excretion of n -type animal, respectively; N_j is the nitrogen amount of j types animal manure; W_n and α_n are the output and nitrogen content of n -type animal manure, respectively.

The number of raised livestock and poultry was determined by the average feeding period combined with the amount of output and inventory, which were mainly derived from China Rural Statistical Yearbook (2010–2019) [33]. The amount of output was taken as breeding quantity if the breeding period was less than one year. For this case, inventory was not considered and the feeding period was the average feeding days. However, the amount of inventory was taken as breeding quantity for livestock and poultry whose average feeding period was more than one year.

2.4. Urban and rural household garbage and nitrogen contents

Urban and rural household garbage (HG) included urban household garbage and rural household garbage. The biodegradable organic part mainly included kitchen waste, waste paper and wood chips, accounting for 66.4% of household garbage in China [24]. They can be buried and returned to the land, which are calculated as follows:

$$TG = (UG + RG) \times (1 - w) = (UG + RP \times DG \times 365) \times (1 - w) \quad (5)$$

$$TN = TG \times o \times \alpha \quad (6)$$

where TG is the total output of urban and rural household garbage each year; UG is the output of urban household garbage; RG is the output of rural household garbage; w is water content; RP is the number of rural population; DG is the per capita daily garbage production; TN is the total organic nitrogen amount of urban and rural household garbage; o is 66.4%, which is the biodegradable organic proportion of household garbage; α is the nitrogen content.

The output data of HG and the number of rural populations were from China Statistical Yearbook (2010–2019) [3]. The per capita daily garbage production was 0.4 kg d^{-1} [34]; the water content was 28.9%, and the nitrogen content (dry weight) was 0.32% [35].

2.5. Municipal sludge and nitrogen contents

The output data of municipal sludge (MS) were obtained from China Urban and Rural Construction Statistical Yearbook (2009–2018) [36], and nitrogen content (dry weight) was 4.17% [35]. The dry matter mass and nitrogen amount of MS are calculated as follows:

$$TS = US + CS \quad (7)$$

$$TN = TS \times \alpha \quad (8)$$

where TS is the total output of MS that year; US is the output of urban sludge; CS is the county sludge. TN is the total nitrogen amount of MS; α is the nitrogen content.

2.6. Other biomass and nitrogen contents

(1) Slaughterhouse waste

Slaughterhouse waste (SW) mainly includes gastrointestinal mucous membrane residue, blood and water as well as discarded fur, hooves and shells, etc. The calculations of SW of raised animals are as follows:

$$TL = S \times W \quad (9)$$

$$N = TL \times p / \alpha \quad (10)$$

where TL is the total dry matter mass of SW; S is the slaughter amount; W is the dry matter mass of each animal waste. N is the nitrogen amount of SW; p is the dry matter crude protein content ratio; α is the protein and nitrogen conversion coefficient.

Slaughter amount (690 million) in 2018 was derived in the relevant industrial reports [37]. For each pig, the dry matter mass waste is 796 g, including 400 g blood, 80 g oil residue, 16 g gastrointestinal mucous membrane residue, 250 g waste hair and 50 g hoof shell [38]. The dry matter crude protein content ratio is 40%; the protein and nitrogen conversion coefficient is 6.25 [39].

(2) Fishery waste

Fishery waste (FW) mainly includes fish waste (scales, skin, bones, head, spines, etc.), shrimp and crab waste. In 2018, the output of fish was 35.57 million t [40], where the fish waste accounted for about 40% [41], with a water content being 74.26% [42], and the nitrogen content (dry weight) 9.39% [43]; The output of shrimp and crab was 7.38 million t [40], the shrimp and crab waste accounted for about 20% [44]. Its water content was 10.21% and nitrogen content (dry weight) 5.73% [45]. The dry matter mass and nitrogen amount of FW are calculated as follows:

$$TF = DF + DS = F \times t_f \times (1 - w_f) + S \times s_f \times (1 - w_s) \quad (11)$$

$$TN = DF \times \alpha_f + DS \times \alpha_s \quad (12)$$

where TF is the total dry matter mass of FW that year; DF and DS are the dry matter mass of fish waste, shrimp and crab waste respectively; F and

S are the fish production, shrimp and crab production respectively; t_f is the fish waste accounts for the proportion of fish production; s_f is shrimp and crab waste accounts for the proportion of shrimp and crab production; w_f is the water content of fish waste; w_s is the water content of shrimp and crabs waste. TN is the nitrogen amount of FW; α_f is the nitrogen content of fish waste (dry matter); α_s is the nitrogen content of shrimp and crab waste (dry matter).

- (3) Wastes from vegetable, orchard and garden; vinasse; vinegar and soy sauce lees; Chinese herbal dreg and mushroom stick

The resources and nitrogen content data of vegetable waste (VW), orchard waste (OW), garden waste (GW), vinasse, vinegar and soy sauce lees (VSL), mushroom stick (MRS), and Chinese herbal dreg (CHD) were obtained through literatures. Water and nitrogen content of the branches of the three most planted fruit trees in China, apple, pear and peach, were taken as their average values. The dry matter mass and nitrogen amount of these biomasses were calculated as follows:

$$DM = FR \times (1 - w) \quad (13)$$

$$N = DM \times \alpha \quad (14)$$

where DM is the biomass dry matter mass that year; FR is the fresh weight of the biomass, w is water content, N is the biomass nitrogen amount; α is the nitrogen content.

2.7. The total biomass and total nitrogen reservoir

The dry matter mass and nitrogen of the above mentioned biomass resources were added together to obtain the total biomass and total nitrogen reservoir.

2.8. Statistical analysis

Microsoft Excel 2010 was used to conduct statistics and process the basic data of various biomass resources. Figures were generated by using Sigmaplot 10.0 (Systat Software Inc., San Jose, CA, USA).

3. Results

3.1. Biomass of main crop straws and their nitrogen reservoir

The straw coefficient and nitrogen content of main crops in China are shown in Table 2, with the yields of different crops being given in Schedule 1. It was found that the straw yields increased sharply from 2009 to 2015, with the highest value reaching to 972.83 million $t \cdot a^{-1}$, then fell a little bit to 964.66 million $t \cdot a^{-1}$ in 2018 (Fig. 1A) after calculated by formula (1). Among them, the maize straw had the highest biomass, accounting for about 50%, followed by wheat, rice and oil

Table 2

The straw coefficient and nitrogen content of main crops in China. The main crops contain maize, wheat, rice, beans, tubers, cotton, oil-bearing, sugar and fiber crops.

Crop	Straw coefficient [47]	Nitrogen content in straws (% dry weight) [35]
Maize	2.00	0.92
Wheat	1.336	0.65
Rice	0.623	0.91
Beans	1.50	2.10
Tubers	0.50	2.51
Cotton	3.00	0.85
Oil-bearing crops	2.00	1.25
Sugar crops	0.10	1.00
Fiber crops	2.50	1.25

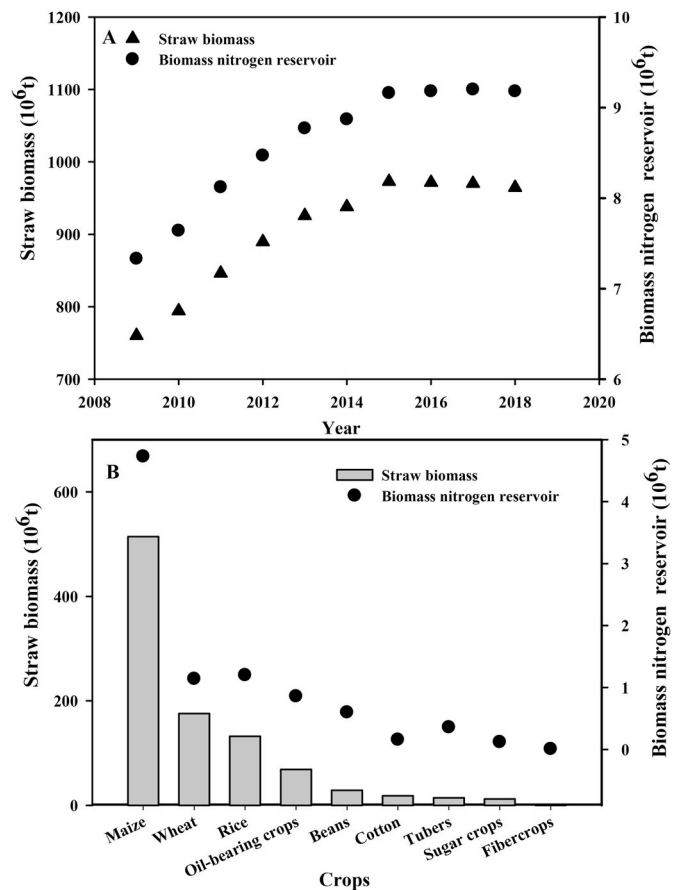


Fig. 1. Straw biomass and nitrogen reservoir of main crops in China from 2009 to 2018 (A) and straw biomass and nitrogen reservoir of different crops in 2018 (B).

crops. Crop straw (CS) contained plentiful nutrients, especially nitrogen, which was the most critical element for crop growth. The nitrogen reservoir in CS of China was about 7.33 million t in 2009, which reached to 9.18 million t in 2018 (Fig. 1A) after calculated by formula (2). Among the CS nitrogen resources, maize straw contributed the most, with its nitrogen amount being significantly higher than others (Fig. 1B).

In China, there were five approaches for CS utilization, i.e., fertilizer, feed, fuel, raw material and base material, accounting for 47.3%, 19.4%, 12.7%, 2.3% and 1.9%, respectively. The rest 16.4% were not utilized or abandoned on the ground [46] (Fig. 2). If the unused CS were also used as fertilizer, the proportion of straw fertilizer would have reached to 63.7%. Thus, the amount of CS used for fertilizer in China might be 614.49 million $t \cdot a^{-1}$, with the nitrogen reservoir being 5.85 million $t \cdot a^{-1}$ in 2018. So there is still huge potential for straws acting as organic fertilizers.

3.2. Manure and urine resources of livestock and poultry and their nitrogen reservoir

The daily excretive manure and urine of various livestock and poultry (LPMU) in China and their corresponding nitrogen reservoir are shown in Table 3. And the amount of output and inventory of all kinds of livestock are provided in Schedule 2. After calculated by formula (3) and (4), it was noted that, from 2009 to 2015, the yield of LPMU in China and the total nitrogen reservoir displayed an increasing trend, but decreased from 2016 to 2018 (Fig. 3A). In 2018, the dry mass of manure was 586.02 million $t \cdot a^{-1}$, the fresh weight of urine output was 878.75 million $t \cdot a^{-1}$. Together, they contained nitrogen amount as large as 13.91 million $t \cdot a^{-1}$. Among them, cattle manure had the highest amount

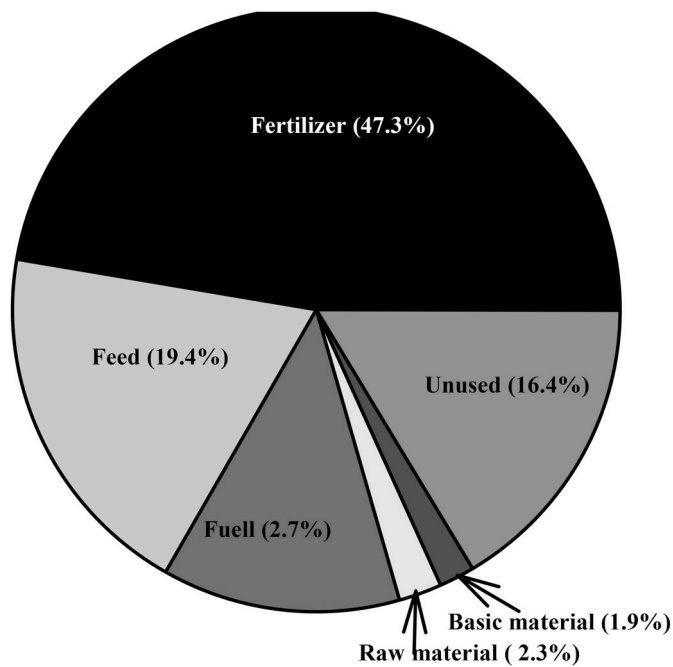


Fig. 2. Proportion of approaches for crop straw utilizations in China.

of nitrogen (4.90 million $t \cdot a^{-1}$), followed by pig (4.03 million $t \cdot a^{-1}$), sheep (2.43 million $t \cdot a^{-1}$) and poultry (2.33 million $t \cdot a^{-1}$). The lowest were found in horses, donkeys, mules and rabbits ($\leq 100,000 t \cdot a^{-1}$) (Fig. 3B).

3.3. Urban and rural household garbage and its nitrogen reservoir

With the fast development of urbanization, the rural population has been surging into cities and the urban population has been increasing year by year. As a result, the amount of urban household garbage in China has gradually increased, while the amount of rural household garbage has decreased from 2009 to 2018 after calculated by formula (5). The total amount of urban and rural waste reached to 220.67 million $t \cdot a^{-1}$ in 2018, with the available organic part in garbage being 146.52 million $t \cdot a^{-1}$. Similarly, the amount of nitrogen has elevated year by year, reaching to 470,341 $t \cdot a^{-1}$ (Fig. 4) in 2018 after calculated by formula (6).

3.4. Municipal sludge biomass and its nitrogen reservoir

Municipal sludge (MS) includes urban sludge and county sludge, with the production being given in Schedule 3. The biomass and nitrogen reservoir of MS in China generally displayed a trend of increasing

from 2011 to 2018 after calculated by Formula (7) and (8). The annual output peaked in 2018, reaching to 13.53 million $t \cdot a^{-1}$, containing nitrogen amount as large as 564,300 $t \cdot a^{-1}$ (Fig. 5).

3.5. Other biomass resources and their nitrogen reservoir

Other biomass resources included waste vegetable waste (VW), orchard waste (OW), garden waste (GW), vinasse, vinegar and soy sauce lees (VSL), mushroom stick (MRS), Chinese herbal dreg (CHD) etc. (Table 4). This part of discarded biomass resource data is obtained by referring to the data in relevant literatures. It was found that among them, the amount of VW resources and nitrogen were the highest, being 269 million $t \cdot a^{-1}$ and 2.394 million $t \cdot a^{-1}$, respectively, followed by GW, vinasse and OW, etc. Comparatively, vinegar lees and slaughterhouse waste (SW) had the least amount of biomass resources and nitrogen by referring to the relevant literatures. These resources have a total biomass of 166.80 million $t \cdot a^{-1}$ with a nitrogen amount of 4.739 million $t \cdot a^{-1}$ (calculated methods refer to formula 9-14).

3.6. The total biomass and nitrogen reservoir

The total dry matter mass of biomass resources available from agriculture was 1.527 billion $t \cdot a^{-1}$ (Fig. 6A), and the urine of livestock (wet weight) was 878.75 million $t \cdot a^{-1}$ (Fig. 3A) in 2018. The total biomass nitrogen reservoir in China was as large as 25.53 million $t \cdot a^{-1}$ (Fig. 6A). Among all, the proportion of nitrogen in LPMU was the highest, accounting for 54.5%; followed by CS (22.9%). The biomass of vegetable waste was lower than HG, however its nitrogen in the former (9.4%) was higher than the latter (1.8%). In addition, VSL occupied 2.3%, followed by MS (2.2%), GW (2.1%), OW (1.8%), FW (1.6%), CHD (0.64%), MRS (0.63%), SW accounted for the least, only 0.13%. In 2018, the amount of SNF used in China was 20.65 million $t \cdot a^{-1}$ (Fig. 6B). However, the actual nitrogen absorbed by agriculture plants was only 30% [4]. According to this judgment, the amount of nitrogen actually absorbed by crops was 6.20 million $t \cdot a^{-1}$. So, we finally found that the total biomass nitrogen reservoir was 4.12 times as large as the amount of the absorbed chemical nitrogen by agriculture plants in China in 2018.

4. Discussion

4.1. Nitrogen resource from crop straws

The excessive application of chemical fertilizer, the low nitrogen use efficiency and the increasingly serious environment pollution have restricted the sustainable development of agriculture worldwide [66]. Rural and urban ecosystems are rich in biodegradable biomass resources with great nutrient potential, which is of great significance to replace chemical fertilizers. As the country with the most abundant straw resources in the world, China's straw biomass is generally growing with the continuous improvement of the comprehensive agricultural

Table 3

Daily excretive manure and urine of livestock and poultry, water content of manure, nitrogen content of manure and urine, and the breeding period of livestock and poultry in China. The livestock and poultry manure were mainly from husbandry industries of pig, cattle, beef cattle, cow, sheep, horse, donkey, mule, rabbit and poultry.

Livestock/ Poultry	Breeding period (d) [26]	Daily excretive manure ($kg \cdot d^{-1}$) [26]	Water content of manure (%) [35]	Nitrogen content of manure (dry weight) (%) [35]	Daily excretive urine ($kg \cdot d^{-1}$) [26]	Nitrogen content of urine (fresh weight) (%) [26]
Pig	199	3.6	68.74	2.09	3.4	0.166
Beef cattle	365	19.9	75	1.67	9.2	0.501
Cow	365	43.4	75	1.67	13.3	0.501
Draft cattle	365	26.6	75	1.67	14.3	0.501
Sheep	365	1.9	50.7	2.01	0.6	0.592
Horse	365	14.2	68.5	1.48	1.9	0.689
Donkey	365	11.8	61.5	0.97	2.9	0.71
Mule	365	11.8	62.9	0.96	2.9	0.71
Rabbit	90	0.2	57.4	2.11	–	–
Poultry	210	0.1	55	1.88	–	–

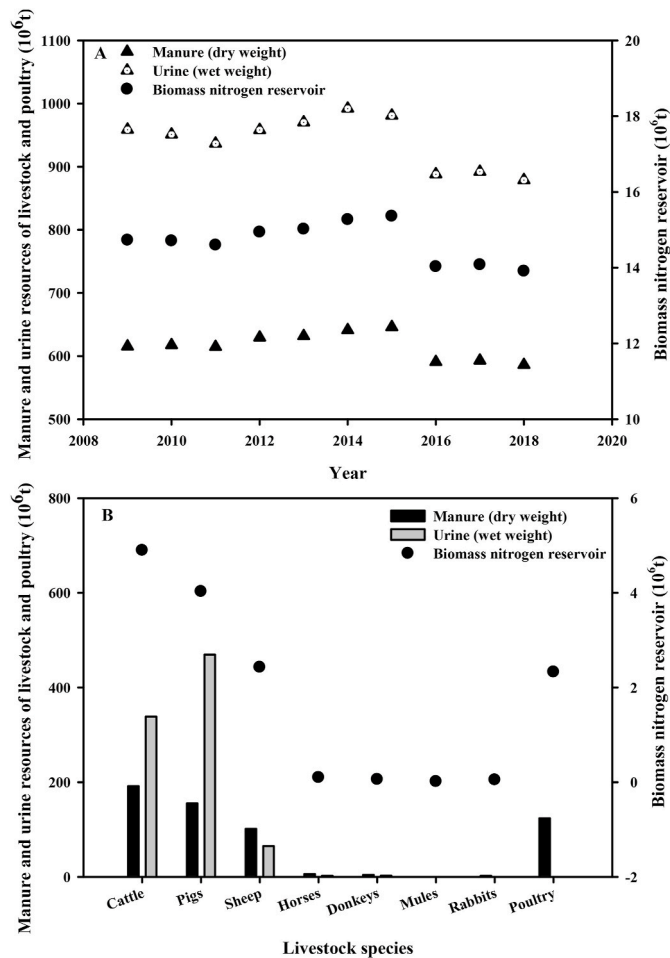


Fig. 3. China's livestock, poultry manure and urine resources and nitrogen reservoir from 2009 to 2018 (A) and the different livestock and poultry manure resources and nitrogen reservoir in 2018 (B).

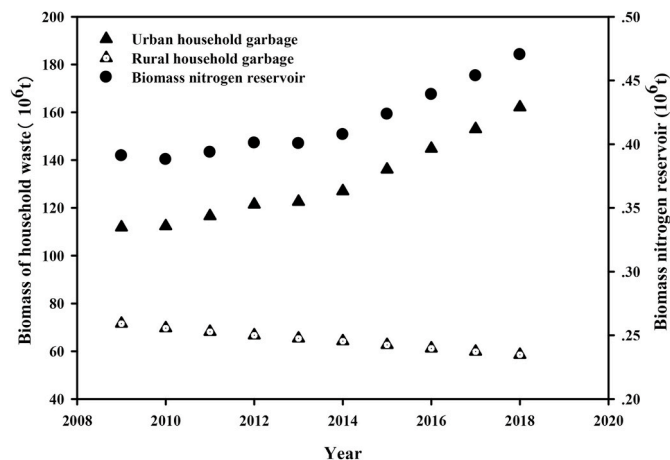


Fig. 4. Biomass and nitrogen reservoir of urban and rural household garbage in China from 2009 to 2018.

production [67]. However, it is not clear how many biomass resources and biomass nitrogen there are in China. Although some investigators estimated the amount of straw resources in different years, the results had certain deviations. For instance, some reported that the biomass of crop straw (CS) in 2006, 2011 and 2015 were 654 [68], 778 [69], 930 million t·a⁻¹ [70], respectively. In this study, we found that the CS

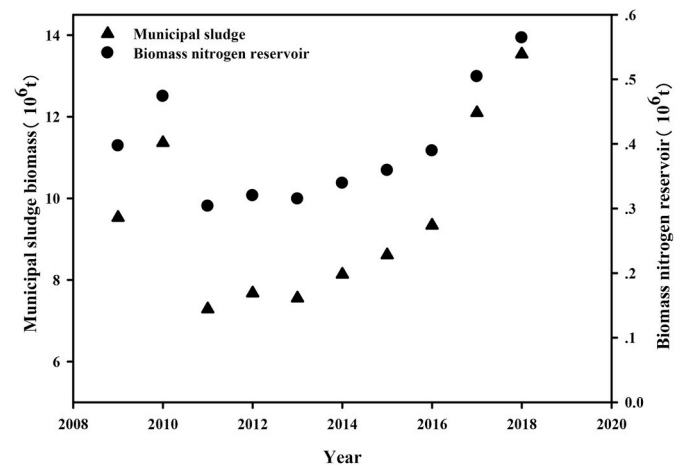


Fig. 5. Biomass and nitrogen reservoir of municipal sludge in China from 2009 to 2018.

biomass in 2011 and 2015 were 846.11 and 972.83 million t·a⁻¹, respectively (Fig. 1A). The differences might be owing to the different sampling methods and or the statistical data sources gained by different researchers. Further, the research object or scope, even definition of CS, and the selection of straw coefficient were also inconsistent. Our data was derived from China Statistical Yearbook, and the straw coefficients were referred to "Assessment of Biomass Resources Availability in China" [47].

CS contains certain nitrogen which originally comes from the air, plays important roles in maintaining nitrogen balance. Studies confirmed that the mineralization process of straw promoted the circulation of soil nitrogen and improved the biological availability of nitrogen [71]. Some found that when all the CSs were returned to the fields, the biomass nitrogen could replace 28.8% of the total synthetic nitrogen fertilizer (SNF) consumption [16]. Chai et al. [72] estimated that the annual average nitrogen nutrient resources of rice, wheat and corn stalks in China for the five years 2013–2017 were 2.09 million t, 1.08 million t and 3.56 million t, respectively. And the substitutable nitrogen fertilizer amounts per unit cultivated land area were 33.6, 23.4 and 51.2 kg hm⁻², respectively [72]. Compared with the above research, our research object was more comprehensive. For instance, the former researchers only limited to the three major grain crops (wheat, rice and corn), here we also included 6 other crops such as cotton, beans and oil-bearing crops, tubers, sugar crops, fiber crops.

In this study, the CS production in China was 964.66 million t·a⁻¹, containing nitrogen 9.18 million t·a⁻¹ in 2018 (Fig. 1A). In that year, the application amount of SNF in China was 20.65 million t·a⁻¹, however, the amount of plant absorption for nitrogen was only 6.20 million t·a⁻¹. So the amount of straw nitrogen was 1.48 times of plant absorption for SCF. Some straws were used as feed, for this reason, livestock and poultry manure contained part of straw resources. To avoid double calculation, we selected the "fertilized" and unused parts of CS as the resources available for fertilizer utilization. Thus, the amount of CS could be used as fertilizers in China might be 614.49 million t·a⁻¹, with a nitrogen reservoir being 5.85 million t·a⁻¹. Therefore, along with the increasing comprehensive utilization rate of CS, the utilization space of such bio-resource is very large in China. One of the potential measures is to develop new straw decomposition technology and make full use of straw nutrients to minimize the chemical fertilizer application.

Nevertheless, there are still some problems such as slow decomposition rate and delayed nutrients release when straws are directly returned to the fields [73]. It is generally believed that the C/N ratio of straw returned to the field should be adjusted to (25–30): 1, which could meet the requirement of microorganisms for nitrogen in the decomposition process. In addition, the recycling of "straws - feed - livestock -

Table 4

Biomass resources and nitrogen reservoir from wastes of vegetable, orchard, garden, slaughterhouse, fish, shrimp and crab shells, vinasse, soy sauce lees, vinegar lees, mushroom stick and Chinese herbal dreg.

Other biomass	Fresh weight (10^6 t a^{-1})	Water content (%)	Dry weight (10^6 t a^{-1})	Nitrogen content (%)	Nitrogen amount (10^6 t a^{-1})	References
Vegetable waste	269	80	53.80	4.45	2.394	[48]
Orchard waste	46.5	18.86	37.73	1.20	0.453	[49–52]
Garden waste	40	40.1	23.96	2.23	0.534	[53–55]
Vinasse	51.4	70	15.42	3.08	0.475	[35,56,57]
Soy sauce lees	11.56	75	2.89	2.92	0.084	[35,58,59]
Vinegar lees	3.2	70	0.96	1.98	0.019	[35,60]
Mushroom stick	45.7	65	16.00	1.01	0.162	[61,62]
Chinese herbal dregs	30	65	10.50	1.55	0.163	[35,63–65]
Slaughterhouse waste (pig)	1.74	31.6	0.55	6.40	0.035	[37–39]
Fish waste	14.23	74.26	3.66	9.39	0.344	[40–43]
Shrimp and crab shells waste	1.48	10.21	1.33	5.73	0.076	[40,44,45]
Sum	–	–	166.80	–	4.739	–

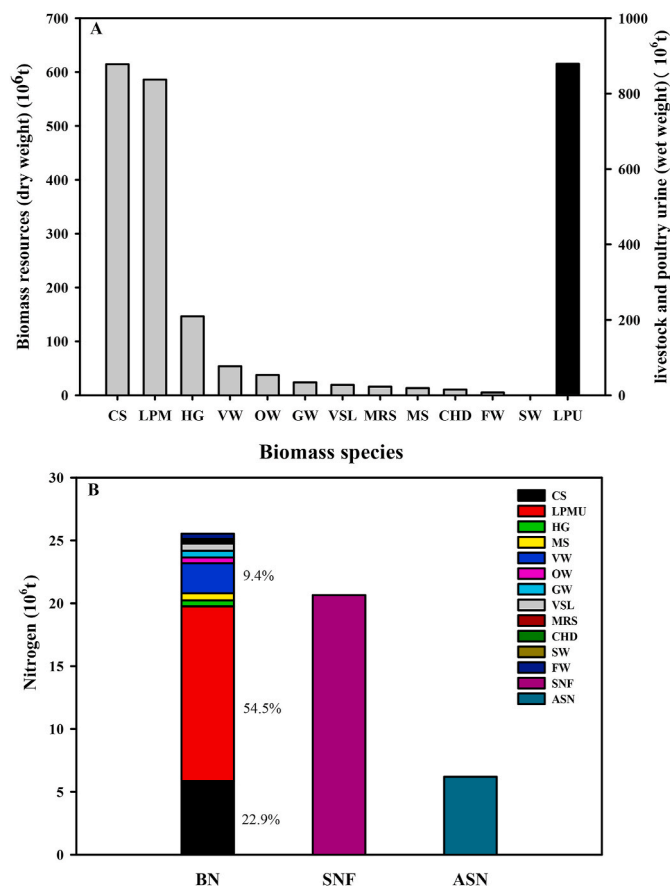


Fig. 6. Biomass resources (A) and total nitrogen amount (B) in China in 2018. CS, LPMU, HG, VW, OW, GW, VSL, MRS, MS, CHD, FW, SW, LPU represent the crop straw, livestock and poultry manure, urban and rural household garbage, vegetable waste, orchard waste, garden waste, vinasse, sauce and soy vinegar lees, mushroom stick, municipal sludge, Chinese herbal dreg, fishery waste, slaughterhouse waste, livestock and poultry urine, respectively. LPMU is the manure and urine of livestock and poultry, BN is biomass nitrogen, SNF is synthetic nitrogen fertilizer, ASN is the absorbed synthetic nitrogen by agriculture plants.

fertilizer-food” model has been suggested and practiced so as to return the nutrients to the field through livestock, thus improving the utilization rate of straws [74]. It is also necessary to actively publicize the dangers of burning straw and the benefits of returning straw to the field, formulate the policy of replanting straw to the field, and encourage farmers to use the technology of returning straws to the fields.

4.2. Manure and urine nutrient resources of livestock and poultry

The production of livestock and poultry excrement in China, increasing at an average rate of 2.32% per year, has exceeded that of industrial solid waste and become the main source of environmental pollution [75]. The emission reduction of livestock manure and urine pollutants should not be underestimated, which is crucial to the realization of national energy conservation and emission reduction targets of greenhouse gases. Understanding the annual output of livestock manure and urine is the basis of its reasonable utilization. Some studies have estimated the excrement and urine output of livestock and poultry in various years in China [26,76,77], however the results have some deviations.

According to previous studies, the manure and urine resources of livestock and poultry (LPMU) in 2009, 2011 and 2015 were 3.26, 2.55 and 3.16 billion t a^{-1} , respectively. In this study, we found that in 2009, 2011 and 2015, the LPMU production in China was 2.90, 2.86 and 2.99 billion t a^{-1} , respectively (Schedule 2), with the dry matter mass of manure being 615.18, 614.55, 645.81 million t a^{-1} , respectively (Fig. 3A). The differences in the above results may be due to the differences in years, discharge coefficient selection, feeding period and estimation of livestock and poultry numbers.

LPMU would become a major source of pollution to the environment if not be properly handled. In fact, they contain a lot of nutrients, especially nitrogen [78]. Some found livestock manure compost application could improve soil fertility [79], for instance, composting of manure and straws had a significant effect on soil organic matter content [80]. Theoretically, the LPMU resources in China contained 11.55 million t nitrogen, which was equivalent to 48.26% of the agricultural nitrogen fertilizer application in 2013 [81]. Even some estimated the LPMU resources in China contained 14.78 million t nitrogen in 2015 [26]. In this study, the nitrogen reservoir of LPMU was 13.91 million t, which was 2.24 times that of plant absorption for SNF in 2018 (Fig. 3B). Strengthening the utilization of LPMU resources can not only reduce environmental pollution, but also reduce chemical fertilizer application and restore soil health. The usage of LPMU as a partial substitute for SNF could increase the average yield of China’s major grain crops by 6.8% and increase the nitrogen use efficiency by 10.4% [82]. The main problems may exist in that, heavy metals, antibiotics and hormones may exceed the standard of organic fertilizers. To face these problems, the government needs to further improve the feed formula at the source and prohibit the addition of harmful substances. In addition, there is a certain N nutrient loss in the composting process. Studies have found that, among the different treatments of livestock manure and urine, direct application to farmland is the simplest approach, while there are certain limitations in transferring livestock manure from farm to farmland. The characteristics of livestock manure and urine make it unable to move in a wide range and across regions. Therefore, under the current economic and feasible technical conditions, arable land consumption is still a relatively economic and feasible means of livestock pollutant

treatment. The government should encourage researchers to strengthen the technical research on returning LPMU to the fields, and solve the problems such as the appropriate amount of livestock manure and urine, the treatment and transportation mode of livestock manure urine and the coordination between livestock manure and inorganic fertilizer.

According to statistics, China imported a total of 115.55 million t grain in 2018, and decreased by 11.5% since 2017 [83]. Among the major grain crops, soybean is the highest with a total import of 88.03 million t, followed by maize (3.52 million t), wheat (3.10 million t) and rice (3.08 million t). The main source countries of imports are the United States, Australia, Brazil and Thailand, among which the United States is the largest source of grain imports for China [84]. Soybean and maize are mainly used for livestock feed, with protein content of 44.31% [85] and 10.57% [86] respectively. The nitrogen content was calculated as 5.76% and 1.59% respectively after dividing by the protein and nitrogen conversion coefficient [39]. The total nitrogen amount of the two grains is about 5.13 million $\text{t}\cdot\text{a}^{-1}$ and soybean contribution is very high, 5.07 million $\text{t}\cdot\text{a}^{-1}$. For this particular part, we included them in the whole amount of manure and urine nutrient resources of livestock and poultry in China.

4.3. Degradable urban and rural household garbage resources

Household garbage (HG) is the main component of urban and rural human wastes in China, with others including kitchen waste, wasted paper, plastic, fabric and glass, wood, ash, metal, etc. [87]. Urban and rural household biodegradable garbage contains a large amount of nutrients, such as nitrogen, phosphorus, potassium and other nutrients. Food waste in the garbage is the major component, accounting for about 40% of the HG [88]. Europe produces about 50 million $\text{t}\cdot\text{a}^{-1}$ of kitchen waste, while Japan produces 20 million $\text{t}\cdot\text{a}^{-1}$. Although China is a developing country, the production of food waste has increased from 77.88 million $\text{t}\cdot\text{a}^{-1}$ to 97.31 million $\text{t}\cdot\text{a}^{-1}$, from 2009 to 2016 [89]. At present, the disposal methods of urban and rural refuse in China include incineration, landfill, composting, etc. Under the action of microorganism, degradable urban and rural HG could be decomposed into organic fertilizers [90]. After the concentrated collection and fermentation of these HGs, the organic matter in the wastes will be rapidly decomposed into substances that can be easily absorbed by crops, thus partly replacing the usage of chemical fertilizers. According to a statistics, 150.00 million t organic fertilizers can be produced by treating 140.00 million t wastes with appropriate manure or CS [91]. A research studied the effect of applying garbage fertilizer on peanut yield in tidal soil showed that garbage fertilizer application could improve the content of organic matter, total nitrogen and effective N, P and K in soil, especially the content of trace elements, among which the content of Fe, Mn and Mo increased by 17–38% compared with the no fertilization treatment [92]. By doing so, we would not only process the harmful garbage effectively, but also utilize wastes to improve crop yield.

This study demonstrated that, the amount of urban and rural HG was nearly 220.67 million t in China in 2018, of which 146.52 million t was organic, with the nitrogen reservoir being 470, 341 t (Fig. 4). That accounted for 7.6% of the crop absorption for SNF. Unfortunately, at present, most of the degradable domestic waste is land-filled or incinerated, which leads to serious environmental pollution and waste of nitrogen resources. A complete garbage classification and collection system should be established to strengthen the supervision and management, garbage fertilization technology and advanced treatment equipment.

4.4. Municipal sludge resources

Municipal sludge (MS) is a kind of solid waste separated from sewage discharged from factories and residential lives in the process of waste water treatment. Worldwide, the total amount of dry sludge has reached to 10 million $\text{t}\cdot\text{a}^{-1}$, 6.6 million $\text{t}\cdot\text{a}^{-1}$ and 2.4 million $\text{t}\cdot\text{a}^{-1}$ in the United

States, European countries and Japan, respectively. In this study, we found that the production of MS in China increased year by year from 2009 to 2018, reached to 13.53 million $\text{t}\cdot\text{a}^{-1}$ in the year of 2018 (Fig. 5). Practically, the environmental problems from treating MS could be solved by recycling it as organic fertilizer. At present, MS has been widely used as an organic fertilizer; as the innocuously treated sludge was rich in nitrogen, phosphorus, potassium and organic matter, whose content is higher than the commonly used farm manure. It also plays an important role in improving the physical and chemical properties of soil and promoting the increase of agricultural and forestry crops [93]. A 20 years sludge application experiment showed that after applying 224 $\text{t}\cdot\text{hm}^{-2}$ sludge, the total C of the farmland soil increased from 1.9% to 4.8%, the total N increased from 0.14% to 0.41%, pH values increased from 6.0 to 7.6, while the total C and N remained basically unchanged in the treatment without sludge [94].

Here in this study, the nitrogen reservoir of MS in 2018 was 564,300 $\text{t}\cdot\text{a}^{-1}$ (Fig. 5), accounting for 9.1% of crop absorption for SNF. If sludge does not contain heavy metals, or below the safe levels required by the State, it can be utilized for growing crops, such as wheat, rice, barley, oats, rye, corn, vegetables, green forage, and fruit trees [95]. However, MS may contain certain heavy metals, if they exceeded the standard levels, sludge cannot be used to produce food, while suitable for producing bio-energy. At present, the common sludge treatment and disposal methods include landfill, incineration, composting, building materials, etc. Sludge compost is one of the important ways of harmless and recycling of MS. Nevertheless, there may exist heavy metals, toxic organic substances and other harmful substances in the sludge [96]. So, we suggested that the government should formulate relevant standards to prevent heavy metals and other harmful substances in sludge exceeding the standard.

4.5. Other biomass resources

With the continuous expansion of vegetable planting area in China, the total output of vegetables and the requirements for higher quality vegetables, the production of vegetable waste (VW) has also sharply increased. The total amount of VW reached to 269 million $\text{t}\cdot\text{a}^{-1}$, or the dry matter mass was 53.80 million $\text{t}\cdot\text{a}^{-1}$ (Table 4). VW was rich in organic matter and various nutrient elements, which can be turned into treasure after harmless treatment [97]. Studies found that VW composting could significantly reduce soil pH and soil bulk density, while increase soil organic matter content, cation exchange capacity and soil porosity [98]. Compared with commercial organic fertilizer, VW composting increased the yield of small rapeseed by 7.0%–15.9% [99]. The nitrogen reservoir in VW was 2.394 million $\text{t}\cdot\text{a}^{-1}$ (Table 4), being equivalence to 38.61% of crop absorption for SNF in 2018.

From 1980 to 2018, the area of orchard in China increased from 1.783 million hm^2 to 11.88 million hm^2 [3]. For normal fruit tree management, most trees need to be pruned every year, resulting in a huge of residual branches. Every year, 100–150 $\text{kg}\cdot\text{tree}^{-1}$, 400–600 $\text{kg}\cdot\text{tree}^{-1}$ and 400–500 $\text{kg}\cdot\text{tree}^{-1}$ branches were cut from apple orchard, peach and pear garden, respectively. The amount of orchard waste (OW) was 46.5 million $\text{t}\cdot\text{a}^{-1}$ and the dry matter mass was 37.73 million $\text{t}\cdot\text{a}^{-1}$, containing pure nitrogen 453,000 $\text{t}\cdot\text{a}^{-1}$ (Table 4), accounting for 7.31% of crop absorption for SNF in 2018. So the pruned and discarded branches of the orchard are valuable and easily gained organic fertilizer sources [100]. They can be pulverized and directly applied to the field, or be composted into biological organic fertilizers.

In China, more than 40 million $\text{t}\cdot\text{a}^{-1}$ urban garden wastes (GWs) are produced, with a dry matter quality of about 23.96 million $\text{t}\cdot\text{a}^{-1}$ and the nitrogen amount 534,000 $\text{t}\cdot\text{a}^{-1}$ (Table 4), accounting for 8.61% of crop absorption for SNF in 2018. Studies have found that GW was rich in nutrient elements, which played a positive role in improving soil fertility and soil physical structure, conserving water, promoting plant growth, and alleviating groundwater pollution [101]. Lignin and cellulose are the main ingredients of GW which are difficult to be degraded. However,

some investigators found that GW combined with furfural residue, bamboo juice liquid, or carbon efficient microbial agents could be decomposed more easily, so as to improve the quality of compost product [102].

China is the world biggest producer of wine, soy sauce, vinegar, edible fungus and Chinese herbal medicine, thus produced a large amount of wastes every year. The resource amounts were 51.4, 11.56, 3.2, 45.7 and 30 million $\text{t}\cdot\text{a}^{-1}$, respectively for vinasse, vinegar lees, soy sauce lees, mushroom stick (MRS), Chinese herbal dreg (CHD). Such biomass resources also have rich nutrients, with a total of 0.903 million $\text{t}\cdot\text{a}^{-1}$ nitrogen (Table 4), accounting for 14.56% of agriculture plant absorption for SNF in 2018. It was noted that vinasse was very beneficial to the growth of wheat, and could promote the secondary rooting and growth of wheat [103]. Compared with composted farmyard manure, the application of mushroom stick could increase the yield of wheat, corn and soybean by 28.7%, 10.5% and 30.6%, respectively [104].

A large amount of slaughterhouse waste (SW) (1.74 million $\text{t}\cdot\text{a}^{-1}$) and fishery waste (FW) (15.71 million $\text{t}\cdot\text{a}^{-1}$) were also produced in China every year (Table 4), which would seriously affect the surrounding environment if not be properly handled. Some reported that SW compost could increase the number of soil microorganisms, improve soil organic matter and add nutrients to the soil [105]. We found that the nitrogen amount in SW was 35,000 $\text{t}\cdot\text{a}^{-1}$ (Table 4), accounting for 0.56% of plant absorption for SNF in China. In addition, the processing of aquatic products also yielded a large number of organic wastes, such as the head, shell and tail of shrimp and crab, the skin, scales, bones, spines and viscera of fishes. Those wastes were also rich in nutrients especially nitrogen [106]. The nitrogen amount available in such reservoir was about 420,000 $\text{t}\cdot\text{a}^{-1}$ (Table 4), which could potentially replace 6.77% of agricultural plant absorption for SNF in China. The government should adopt policies to strengthen the key technologies research on returning the resources to farmland after harmless treatment.

4.6. The practical values, questions and suggestions

The biomass and nitrogen resources in China were systematically described for the first time. These biomass produced around agriculture, urban and rural life has a nitrogen reservoir of 25.53 million t (Fig. 6B), which was 4.12 times that of plants absorption for SNF in China. In this case, chemical fertilizer plants are not needed in China if these biodegradable biomass resources can be fully utilized as organic fertilizers. There are also many other available biomass resources, such as forest litter and waste grassland, even the sea organic wastes are rich in nutrients. These resources were not considered in this paper because of their important roles in natural ecosystems. Theoretically, all the above biomass resources can be used as fertilizers to enrich the farmland and increase agriculture yields. If those bio-resources are largely used in through effective technological means, it would be conducive to reduce or even eliminate the use of chemical fertilizers. Thus our findings might have a guiding significance for promoting environmental protection, recycling natural resources and realizing sustainable development of agriculture.

At present, there are still some challenges in the utilization of these biomass resources in China. Firstly, the technical means of resource utilization are not mature enough, leading to low resource utilization rate. Secondly, it might lead to other environmental pollution, such as

heavy metals in household garbage and sludge. Thirdly, there is imperfection for the policies and regulations. For instance, although China has issued a number of relevant laws, regulations and policies of poultry and animal feces life garbage disposal of straw, they only expounded the macroscopical method and the overall goal of the behavior of bio-resource products. There are not standard specifications in details, which led to different levels of recycling, while some recycling products might also have security risks.

Therefore, we suggest that the government should pay more attention to the utilization of bio-resources in the future, formulating a series of laws and policies and improving citizens' awareness of resource utilization. The government should also strengthen the research and application of key technologies for returning bio-resources to land and popularize the existing mature technologies for recycling resources. At the root, the government should actively support environmental protection enterprises and subsidize waste reuse enterprises.

5. Conclusion

In conclusion, China was actually rich in biomass nitrogen resources, with the total amount of the 12 kinds of dry biomass being 1.527 billion $\text{t}\cdot\text{a}^{-1}$ and the urine of animals 878.75 million $\text{t}\cdot\text{a}^{-1}$. The total nitrogen reservoir of these biomass resources was 25.53 million $\text{t}\cdot\text{a}^{-1}$, which was 4.12 times that of synthetic nitrogen fertilizer actually absorbed by agricultural plants in China in 2018. The highest nitrogen contributor came from livestock and poultry manure, followed by crop straw and vegetable waste, with the least coming from slaughterhouse waste. The challenges still exist in developing new compost technologies and innovative mechanism of biomass nitrogen utilization. The government should take effective measures to use the huge biomass resources in all agriculture ecosystems, so as to realize a virtuous cycle of bio-resources and sustainable agricultural development.

CRedit authorship contribution statement

Xiaohui Cui: Data curation, Formal analysis, Investigation, Software, Writing - original draft, Writing - review & editing. **Liye Guo:** Data curation, Formal analysis, Methodology, Writing - review & editing. **Caihong Li:** Formal analysis, Methodology, Software. **Meizhen Liu:** Formal analysis, Methodology. **Guanglei Wu:** Formal analysis, Methodology, Software, Project administration. **Gaoming Jiang:** Conceptualization, Formal analysis, Funding acquisition, Project administration, Writing - review & editing.

Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rser.2020.110215>.

Schedule 1

Main crop and straw yields in China from 2009 to 2018

Crops	Yield (10 ⁶ t)									
	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Maize	173.26	190.75	211.32	229.56	248.45	249.76	264.99	263.63	259.07	257.17
Straw	346.52	381.50	422.63	459.12	496.91	499.53	529.98	527.27	518.14	514.35
Wheat	115.83	116.14	118.62	122.54	123.71	128.32	132.64	133.27	134.33	131.44
Straw	154.75	155.16	158.48	163.71	165.28	171.44	177.21	178.05	179.47	179.56
Rice	196.20	197.23	202.88	206.53	206.28	209.61	212.14	211.09	212.68	212.13
Straw	122.23	122.87	126.39	128.67	128.52	130.59	132.16	131.51	132.49	132.16
Beans	19.05	18.79	18.63	16.81	15.42	15.64	15.12	16.51	18.42	19.20
Straw	28.57	28.08	27.95	25.21	23.14	23.47	22.69	24.76	27.62	28.80
Tubers:	27.93	28.43	29.24	28.83	28.55	27.99	27.29	27.26	27.99	28.65
Straw	13.96	14.21	14.62	14.41	14.28	13.99	13.65	13.63	13.99	14.33
Cotton:	6.24	5.77	6.52	6.61	6.28	6.30	5.91	5.34	5.65	6.10
Straw	18.71	17.31	19.56	19.82	18.85	18.90	17.72	16.03	16.96	18.31
Oil-bearing crops:	31.39	31.57	32.13	32.86	32.87	33.72	33.91	34.00	34.75	34.33
Straw	62.79	63.14	64.25	65.71	65.75	67.44	67.81	68.00	69.50	68.66
Sugar crops:	117.47	113.03	116.63	124.52	125.55	120.89	112.15	117.16	113.79	119.37
Straw	11.75	11.30	11.66	12.45	12.56	12.09	11.22	11.72	11.38	11.94
Fiber crops:	0.32	0.24	0.22	0.20	0.18	0.16	0.16	0.18	0.22	0.20
Straw	0.80	0.61	0.56	0.49	0.44	0.41	0.39	0.45	0.55	0.51

Schedule 2

Numbers of livestock and manure productions (wet weight) in China from 2009 to 2018

Livestock	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Number of pig (10 ⁶)	645.39	666.86	663.26	697.89	715.57	735.10	708.25	700.74	702.02	693.82
Feces (10 ⁶ t)	462.35	477.74	475.160	499.97	512.64	526.623	507.39	502.01	502.93	497.06
Urine (10 ⁶ t)	436.67	451.20	448.76	472.19	484.16	497.37	479.20	474.12	474.99	469.44
Number of beef cattle (10 ⁶)	59.19	67.39	66.46	66.98	68.39	70.41	73.73	61.81	66.18	66.18
Feces (10 ⁶ t)	429.91	489.48	482.76	486.52	496.72	511.42	535.53	448.96	480.69	480.73
Urine (10 ⁶ t)	198.75	226.29	223.19	224.92	229.64	236.43	247.58	207.56	222.23	222.25
Number of cow (10 ⁶)	12.60	14.20	14.40	14.93	14.41	14.99	15.70	10.37	10.79	10.38
Feces (10 ⁶ t)	199.64	224.96	228.14	236.65	228.27	237.47	248.74	164.27	171.05	164.38
Urine (10 ⁶ t)	61.18	68.94	69.91	72.52	69.95	72.77	76.26	50.34	52.42	50.38
Number of draft cattle (10 ⁶)	35.47	24.67	22.74	21.51	21.06	20.38	18.74	16.17	13.41	12.59
Feces (10 ⁶ t)	344.42	239.56	220.78	208.88	204.42	197.87	181.97	156.95	130.29	122.26
Urine (10 ⁶ t)	185.16	128.79	118.69	112.29	109.90	106.37	97.82	84.37	69.99	65.72
Number of horse (10 ⁶)	6.79	6.77	6.71	6.34	6.03	6.04	5.91	5.51	3.44	3.47
Feces (10 ⁶ t)	35.17	35.09	34.77	32.83	31.24	31.32	30.62	18.20	17.81	18.00
Urine (10 ⁶ t)	4.71	4.70	4.65	4.39	4.18	4.19	4.10	2.44	2.38	2.41
Number of donkey (10 ⁶)	6.48	6.39	6.47	6.36	6.03	5.83	5.42	2.59	2.68	2.53
Feces (10 ⁶ t)	27.93	27.55	27.90	27.40	25.99	25.09	23.35	11.17	11.53	10.91
Urine (10 ⁶ t)	6.86	6.77	6.86	6.73	6.39	6.17	5.74	2.74	2.83	2.68
Number of mule (10 ⁶)	2.79	2.70	2.60	2.49	2.30	2.24	2.10	0.85	0.81	0.76
Feces (10 ⁶ t)	12.03	11.62	11.19	10.73	9.92	9.67	9.04	3.64	3.49	3.26
Urine (10 ⁶ t)	2.96	2.85	2.75	2.64	2.44	2.38	2.22	0.89	0.86	0.80
Number of sheep (10 ⁶)	284.52	280.88	282.36	285.04	290.36	303.15	310.99	299.31	302.32	297.14
Feces (10 ⁶ t)	197.32	194.79	195.82	197.68	201.37	210.23	215.68	207.57	209.66	206.06
Urine (10 ⁶ t)	62.31	61.51	61.84	62.42	63.59	66.39	68.11	65.54	66.21	65.07
Number of rabbit (10 ⁶)	432.81	464.53	474.70	487.77	503.67	516.79	523.57	350.57	319.55	316.71
Feces (10 ⁶ t)	7.79	8.36	8.54	8.78	9.06	9.30	9.42	6.31	5.75	5.70
Number of poultry (10 ⁶)	10,610	11,010	11,330	12,080	11,900	11,540	11,990	13,200	13,020	13,090
Feces (10 ⁶ t)	222.81	231.21	237.93	253.68	249.90	242.34	251.79	277.20	273.42	274.89

Schedule 3

Municipal sludge resources in China from 2009 to 2018

Sludge	Production (10 ⁶ t)									
	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Urban sludge	8.93	10.33	6.50	6.55	6.56	7.11	7.46	7.99	10.53	11.76
County sludge	0.60	1.04	0.78	1.12	0.99	1.02	1.15	1.34	1.56	1.77

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